

FENG HE

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EDUCATION

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- M.S. in Neuroscience, University of Science and Technology of China 2023.09 – 2026.06
- Research focus: machine learning, representation learning, and scientific imaging
 - Recipient of the **China National Scholarship** (Top 0.1% nationwide), 2026
- B.S. in Computer Science, Yangtze University 2019.09 – 2023.06
- GPA: 4.07/5.0; **Ranked 1st** overall among ~300 students (graduate recommendation)

RESEARCH INTERESTS

Reliable reasoning and agentic behavior in large language models, particularly how models make decisions under unreliable premises, feedback, and verification signals. I develop controlled evaluations and mitigation methods to understand and improve the reliability of LLM reasoning and multi-step agents.

SELECTED PUBLICATIONS

Full publication list available on [Google Scholar](#).

LLM Reliability, Agents, and Reasoning

1. **Feng He**, Hejia Wang, Linghao Meng, Ming Gao, and Qiankun Li. Verification Trap: Understanding Test-Time Selection Failures under False Premises in Code Generation. *EMNLP 2026, Oral Presentation*.
 - Led the project and identified the Verification Trap, a test-time selection failure mode under false premises, with mechanistic analysis, gold-free prediction, and decoupled-evidence mitigation for code generation.
2. **Feng He**, Hejia Wang, Linghao Meng, Huabao Chen, Mengyao Lyu, and Qiankun Li. Understanding Pre-Scaling Reasoning under False Premises in Code Generation. *ICLR 2027 Submission*.
 - Led the project and introduced Pre-Scaling Reasoning (PSR), a framework for predicting specification-conditioned vulnerability before generation and selectively allocating additional reasoning to high-risk instances.
3. **Feng He**, Zhenting Wang, Dongfang Liu, Ruixiang Tang, and Qiankun Li. HIVE: Understanding Post-Hallucination Reasoning in Vision Language Models. *ECCV 2026*.
 - Analyzed LLM reasoning under controlled semantic perturbations in scientific domains, characterizing entropy shifts, convergence stability, and mechanism-level factors that govern beneficial hallucinations.

Foundation Models for Scientific and Multimodal AI

4. **Feng He**, Guodong Tan, Qiankun Li, Jun Yu, and Quan Wen. From Pixels to Views: Learning Angular-Aware and Physics-Consistent Representations for Light Field Microscopy. *NeurIPS 2025*.
 - Developed physics-consistent 3D representations for biological light-field microscopy using multi-view angular cues.
5. **Feng He**, Hanlin Li, Xin Ning, and Qiankun Li. BeautyDiffusion: Generative Latent Decomposition for Makeup Transfer via Diffusion Models. *Information Fusion*, 2025.
 - Introduced a generative latent decomposition framework enabling disentangled makeup transfer in diffusion models.

RESEARCH EXPERIENCE

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- University of Southern California**, Research Collaborator 2026.06 – Present
Collaborating with Prof. Jieyu Zhao on reliable multi-agent systems. Investigating how oversight, feedback, and information asymmetry shape strategic behavior in multi-stage LLM agent interactions.
- Purdue University**, Research Collaborator 2025.05 – 2026.01
Collaborated with Prof. Dongfang Liu on LLM hallucination and reasoning. This collaboration resulted in a first-author paper at *ECCV 2026*.
- University of Science and Technology of China**, Research Assistant 2023.09 – 2025.05
Conducted research on representation learning and scientific imaging. Developed physics-consistent representation learning methods, resulting in a first-author paper at *NeurIPS 2025*.